

Environmental Sensing on XGS-PON OLT Using Upstream Burst Window Alignment

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Abstract We demonstrate PON integrated sensing and communication (ISAC) for fiber ambient temperature estimation and continuous monitoring. Using operational XGS-PON we measure change in fiber RTT at OLT using burst-mode upstream alignment. Experimental results in a climatic chamber show 0.63°C temperature RMS error over 4-km-long fiber. ©2026 The Author(s)

Introduction

Recently, fiber optic sensing is gaining interest as a method for both environment sensing [1], as well as fiber plant monitoring and protection using existing fiber infrastructure in a cost-efficient and scalable way [2]. It is also being considered for use in passive optical networks (PON) [3-6], which are among the most widely deployed access technologies worldwide.

Currently, sensing is often performed using dedicated hardware, which can be configured in a *backward sensing* manner (implying distributed fiber optic sensing – DFOS, i.e., longitudinally-resolved measurement using backscattered light [2], such as optical time-domain reflectometry – OTDR or distributed acoustic sensing – DAS [3]) or *forward sensing* (most often limited to spatially-integrated sensing, lumping together all changes in the link in a single measurement, such as state of polarization – SOP [4]). A complementary approach not requiring dedicated hardware may exploit the data (downstream or upstream) traffic transported by PON, along with telemetry of built-in performance monitoring [3] or signal quality metrics [7] that may be locally or remotely read using, e.g., Physical Layer Operations, Administration and Maintenance (PLOAM) channel. Telemetry sensing has the advantage of being readily available in all PON system generations, including already deployed systems, such as GPON [8], potentially allowing for retrofitting the sensing functionality via software. Examples include monitoring of fiber temperature through changes in roundtrip time (RTT) via monitoring updates to EqD [8,9] or SNR measurements at the receiver when using DSP-

based receivers (limited to 50G-PON, though) [7,10].

Telemetry-based sensing enhances the intrinsic value of both brown field and green field PON installations by augmenting their primary function of broadband service delivery with seamlessly coexisting sensing, while avoiding additional capex. This PON Integrated Sensing and Communication (ISAC) concept parallels current radio-engineering efforts that leverage cellular networks for simultaneous sensing and data communication [11].

The PON ISAC concept presents fiber broadband operators with a new opportunity to provide environmental temperature data to third parties in a sensing-as-a-service (SaaS) manner, offering improved geographical resolution compared to the standard use of meteorological weather stations. This monitoring data can be valuable for various applications, including smart city initiatives, environmental and public health monitoring, climate research, and fire or pipeline leaks detection and monitoring [12], among others.

Proof-of-concept experiments in GPON [9,13] have shown that by rebooting the ONU, information about environmental temperature could be retrieved from the equalization delay (EqD) parameter [8,9]. EqD is used to compensate for changes in the RTT of the signal, which are affected by temperature-induced drifts in the refractive index of the fiber. Thus, from EqD readings, temperature changes can be estimated. However, the EqD in GPON and later PON generations is quantized in steps of 8 bits at 2.48832 Gbps to reduce the number of Drift of

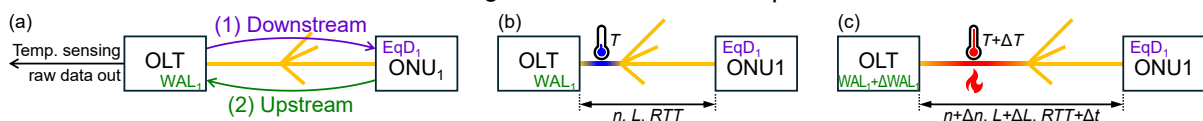


Fig. 1: Schematics outlining PON temperature sensing using EqD and WAL pointer. (a) Location of EqD and WAL parameters. (b,c) Changes occurring to the fiber plant, EqD and WAL pointer parameters as the fiber ambient temperature increases.

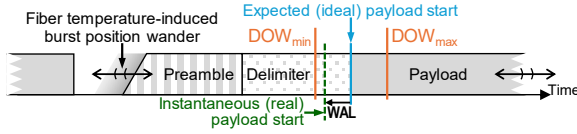


Fig. 2: Illustration of an upstream burst received at the OLT showing WAL pointer and EqD parameters, as well as DOW_{min} and DOW_{max} thresholds.

Window (DOW) alarms. This precludes high-accuracy real-time EqD readings, resulting in poor temperature resolution. Besides, forcing the EqD update by rebooting the ONU causes the interruption of the customer service and thus this approach is not feasible in production environments.

In this paper, we present a new approach that overcomes these limitations. It is shown that estimations derived from the OLT-internal Window Alignment (WAL) pointer in XGS-PON [14] allows for a significant enhancement of temperature resolution, enabling seamless continuous monitoring, without disrupting normal network operation. Experiments conducted in a climatic chamber demonstrate temperature tracking with a significant increase in resolution over previous works, yielding a root-mean-square error (RMSE) of 0.63°C for a fiber length of 4 km.

Method

The new temperature retrieval method is shown in Fig. 1. It represents a link between OLT and one of the ONUs, ONU1, where temperature-induced RTT changes result in drifts of the WAL parameter. In time-domain multiple access (TDMA) PON, each ONU is assigned a specific time slot to send data in the upstream direction in the form of an optical burst, Fig. 2. While the EqD parameter is only calculated by the OLT and sent to the ONUs at the ONU start-up, or when the DOW threshold is crossed (coarse delay), the WAL pointer refers to the instantaneous upstream transmission offset (fine delay), as measured between the allocated start time in the upstream transmission window of an ONU and the actual time of arrival of the burst at the OLT, see Fig. 1(a) and Fig. 2 for details. By combining EqD and WAL pointer readings, even large RTT changes can be measured. Temperature changes along the deployed fibers of the optical distribution network of a PON, especially strong in aerial deployments, result in drifts in the fiber refractive index, mainly due to the thermo-optic coefficient since the thermal expansion coefficient can be neglected [15]. This changes the RTT of the signal and, therefore, the arrival of the data. By monitoring the WAL, one can infer changes in the average temperature along the link between OLT and ONU, without interrupting

normal data transmission or PLOAM overhead, see Fig. 1(b,c). Combining the information from several ONUs and integrating it with a Geographic Information System (GIS), temperature maps can be developed [13].

The temperature changes averaged along the fiber (ΔT) can be estimated from the change in the WAL pointer Δt according to Eq. (1) as

$$\langle \Delta T \rangle = \frac{\Delta t \cdot c}{\frac{dn}{dT} \cdot (n_{down} + n_{up}) \cdot L}, \quad (1)$$

where c is the speed of light in vacuum, dn/dT is the fiber thermo-optic coefficient, n_{down} and n_{up} are the refractive index of the fiber in downstream and upstream, respectively, L is the OLT-ONU fiber length, and Δt is the change in RTT derived from the change of the WAL pointer, ΔWAL , given by

$$\Delta t = \Delta WAL \cdot \frac{FrameDuration}{BitsPerFrame/4}, \quad (2)$$

where *FrameDuration* is the Transmission Convergence (TC) frame duration, i.e. $125 \mu\text{s}$ for ITU-T PON recommendations including XGS-PON, and *BitsPerFrame* is the number of bits in the upstream XGS-PON frame at 9.95328 Gbit/s bitrate. Please note that *BitsPerFrame* is divided by 4 in order to comply with the EqD unit definition in XGS-PON, which assumes 2.48832 Gbit/s line rate, regardless of the actual upstream line rate.

The enhancement in temperature estimation over previous works is mainly due to the twofold enhancement in EqD resolution in XGS-PON compared to GPON and the ability to rapidly read WAL parameter, which avoids the need for constant restarting of the ONU and allows significant oversampling and postprocessing. In actual hardware, the instantaneous burst arrival position at the OLT, Fig. 2, varies not only because of external temperature-induced fiber delay, but also due to the variability of the exact ONU upstream start time, as well as internal parallelized processing of the OLT upstream acquisition system. Counterintuitively, this random instantaneous noise can improve the accuracy of WAL, since a large quantization error due to line-rate-invariant quantization of WAL can be averaged by an additional random error with finer accuracy, effectively acting as a dither, allowing for an enhancement in the estimation via averaging [16].

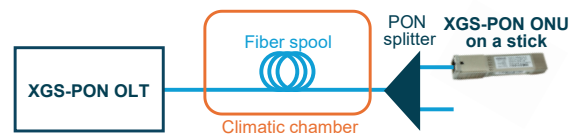


Fig. 3: Experimental setup for temperature monitoring consisting of an XGS-PON system and a climatic chamber.

Experimental results

Proof-of-concept experiments were conducted using a climatic chamber, as shown in Fig. 3. The experimental setup consists of a Nokia OLT chassis and line card, modified for research, equipped with a multi-PON module (MPM) with GPON and XGSPON optics in a PON port, connected to an ITU-T G.652 fiber spool of 4239 m, placed inside the climatic chamber, which emulates changes in the environmental temperature of the fiber network. The output of the fiber coil is connected to a 1×16 optical splitter and then connected to one Nokia XS-010S-Z XGSPON ONU on a stick, receiving around -18 dBm of optical power and with active internet, voice and video services. The temperature in the chamber was changed from 25.5°C to 40°C following a sawtooth pattern with a period of 11 hours and 24 minutes. The XGSPON OLT was used to track the temperature change by reading the WAL pointer every 20 seconds. It produced no impact on customer services running at the ONU. Figure 4 shows a comparison between the temperature readings from the climatic chamber and the estimations from XGSPON where each point is a moving average of 30 samples (10 minutes). The data has been adjusted for a time lag of 25 minutes between PON temperature estimations and chamber readings associated with thermal inertia of the fiber spool, believed to be caused by the insulation provided by the plastic casing and the fiber coiled on the spool. The thermo-optic coefficient used to retrieve temperatures was $7 \cdot 10^{-6} \text{ K}^{-1}$ and the refractive index of both windows has been assumed to be the same 1.467, for the sake of simplicity. There is good agreement between both methods with a RMSE

of 0.63°C for the optimum time lag. As a comparison, similar experiments with GPON have been carried out, showing an RMSE of 1.07°C for the optimum lag correction with the same fiber.

Conclusion

We propose a new approach for estimating changes in the average environmental temperature using operational PON and ODN infrastructure, leveraging the WAL pointer in XGS-PON without causing customer service interruption during monitoring. The WAL pointer parameter is internal to the OLT and has a high refresh rate, avoiding generation of additional PLOAM traffic, and allowing for enhanced resolution for typical PON distances of around 5 km.

This feedforward upstream sensing method operates seamlessly during normal system activity, causing no service interruptions or interference. The results show that the temperature resolution can be enhanced compared to GPON measurements.

Exploiting operational traffic for sensing introduces no additional hardware requirements, opening a pathway for new applications on top of existing PON deployments. Fusing information from multiple ONUs sharing the same feeder section can increase resolution. Temperature resolution may be further enhanced through more accurate methods for measuring the burst arrival time at the OLT and by mandating that the EqD unit be internally accurate to a single bit at the nominal line rate. This approach might be of interest for future standardization efforts, given its potential advantages in the context of sensing-as-a-service in optical networks.

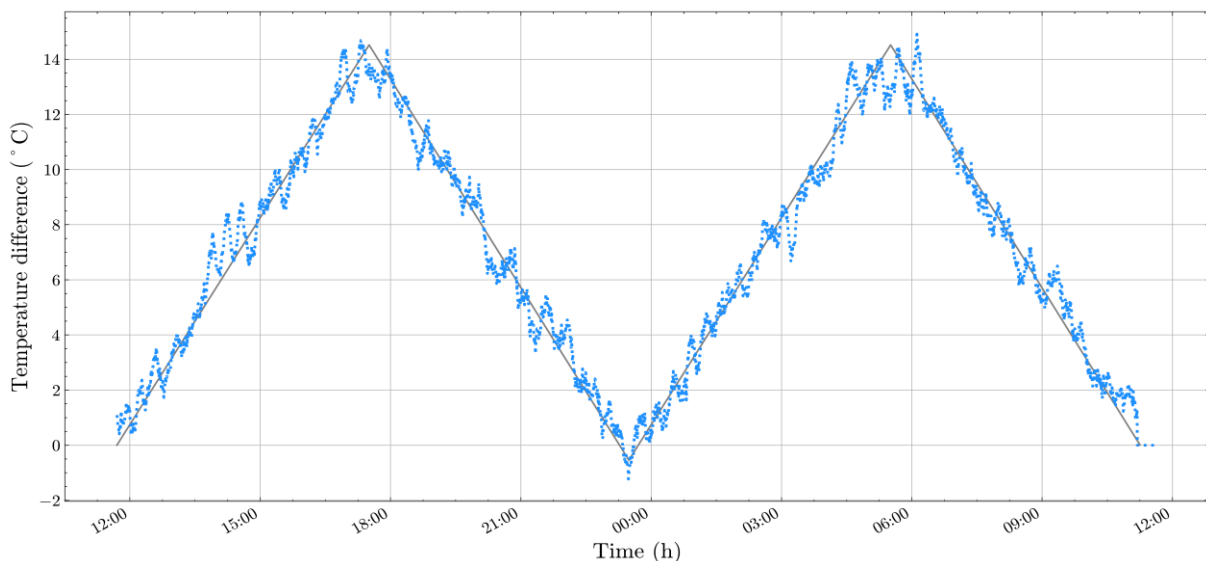


Fig. 4: Temperature difference measured from WAL pointer in XGS-PON with a 4 km fiber spool (blue); temperature change readings from the temperature sensor of the climatic chamber (gray).

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